PROBLEM SET 8

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 30)

1) **(10 pts)** Recall that a particle in a one-dimensional box extending from $x = 0$ to $x = L$ is confined to the region $0 \leq x \leq L$; its wave function vanishes at the edges $x = 0$ and $x = L$ and beyond (see Exercise 5.2.5). Consider now a particle confined in a three-dimensional cubic box of volume L^3. Choosing as the origin one of its corners, and the x, y, and z axes along the three edges meeting there, show that the normalized energy eigenfunctions are

$$\psi_E(x, y, z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_x \pi x}{L}\right) \cdot \sqrt{\frac{2}{L}} \sin\left(\frac{n_y \pi y}{L}\right) \cdot \sqrt{\frac{2}{L}} \sin\left(\frac{n_z \pi z}{L}\right)$$

with energy $E = \dfrac{\hbar^2 \pi^2}{2ML^2}(n_x^2 + n_y^2 + n_z^2)$, and n_i are positive integers.

2) **(10 pts)** Quantize the two-dimensional oscillator for which

$$\mathscr{H} = \frac{p_x^2 + p_y^2}{2m} + \frac{1}{2} m\omega_x^2 x^2 + \frac{1}{2} m\omega_y^2 y^2$$

hint: quantize, i.e., replacing the classical variables with quantum operators; you may refer to general results from section 7.3 (Eq. 7.3.22)

(a) **(5 pts)** Show that the allowed energies are

$$E = (n_x + 1/2)\hbar\omega_x + (n_y + 1/2)\hbar\omega_y, \qquad n_x, n_y = 0,1,2, \ldots$$

(b) **(5 pts)** Write down the corresponding wave functions in terms of single oscillator wave functions. Verify that they have definite parity (even/odd) number $x \to -x, y \to -y$ and that the parity depends only on $n = n_x + n_y$.

3) **(5 pts)** Two identical bosons are found to be in states $|\phi\rangle$ and $|\psi\rangle$. Show that the normalized state vector describing the system when $\langle\phi|\psi\rangle \neq 0$ is

$$|\lambda\rangle = \frac{|\phi\psi\rangle + |\psi\phi\rangle}{\sqrt{2(1 + |\langle\phi|\psi\rangle|^2)}}$$

4) **(5 pts)** When an energy measurement is made on a system of three bosons in a box, the n values obtained were 3, 3, and 4. Write down a symmetrized, normalized state vector $|\psi\rangle$.